UTM
UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 10:
ETHICS IN DATA VISUALIZATION

DATA WRANGLING AND VISUALIZATION (SAIA2143)

COURSE LEARNING OUTCOMES

At the end of this topic, students should be able to:

a. Analyze how visualizations influence perception
b. Evaluate ethical dimensions in data representation
c. Identify and mitigate bias in visualizations
d. Critically assess real-world examples of ethical and unethical data visualizations

TOPIC 10: ETHICS IN DATA VISUALIZATION

a. Introduction to Ethics in Data Visualization

b. Ethics in Practice: Misleading Visual Techniques

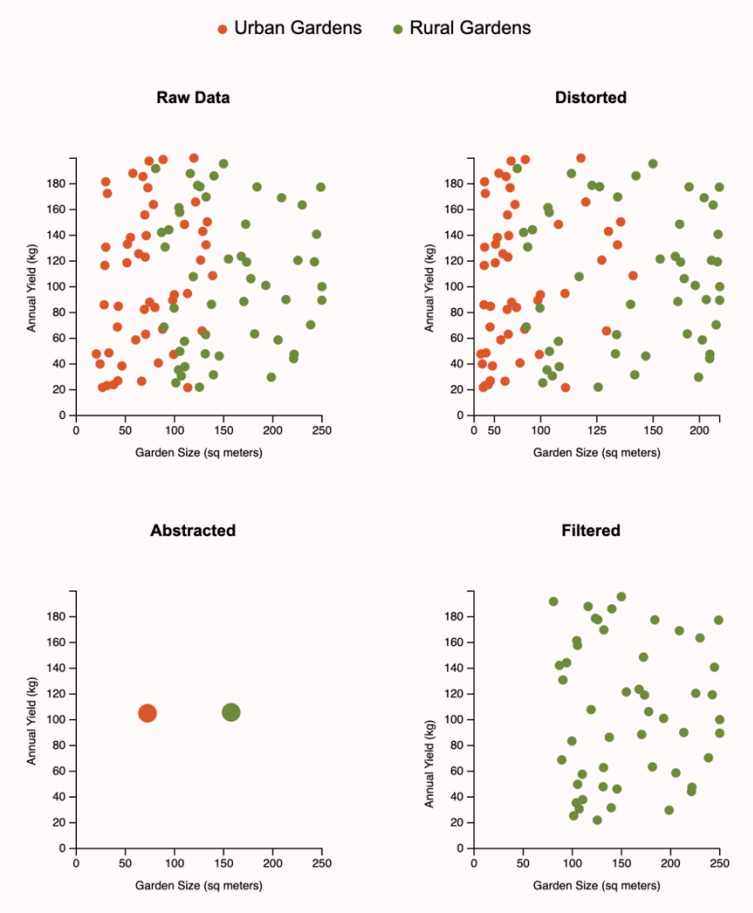
c. Understanding Bias in Visualization

Defining Ethics in Data Visualization

Ethics in data visualization refers to the **responsible use of data and visuals** to communicate information **truthfully, transparently, and fairly**, without misleading the viewer—even unintentionally.

"Just because a chart is factually correct doesn't mean it's ethically sound."

Key Concepts



- **Truthfulness**: Present data without distortion.
- **Clarity**: Ensure the message is not obscured.
- **Fairness**: Avoid visual bias that could mislead interpretation.
- **Accountability**: Consider the impact on users, especially non-experts.

Why Ethics Matter in Data Communication

- When data visualizations are used in business, healthcare, media, and government, **design choices influence public decisions**.

- Ethical design builds **trust** and supports **informed decision-making**.

Why Ethics Matter in Data Communication

Ethical Visualizations

- Promote data literacy
- Maintain objectivity
- Serve the audience's right to understand the data

Unethical Visualizations

- Lead to misinformation or manipulation
- Can result in public harm or distrust
- Prioritize persuasion over truth

Real-World Consequences of Unethical Visualization

Case Study	Description	Ethical Concern
COVID-19 dashboards	Some charts exaggerated death rates or lacked context	Panic, misinterpretation
Election maps	Red/blue maps misrepresented voter distribution	Misleading area vs. population
Product sales graphs	Y-axis starts at non-zero value to exaggerate growth	Distortion of scale
Crime heatmaps	Poor legend or outdated data led to community stigmatization	Data bias and harm

Design Choices That Shape Perception

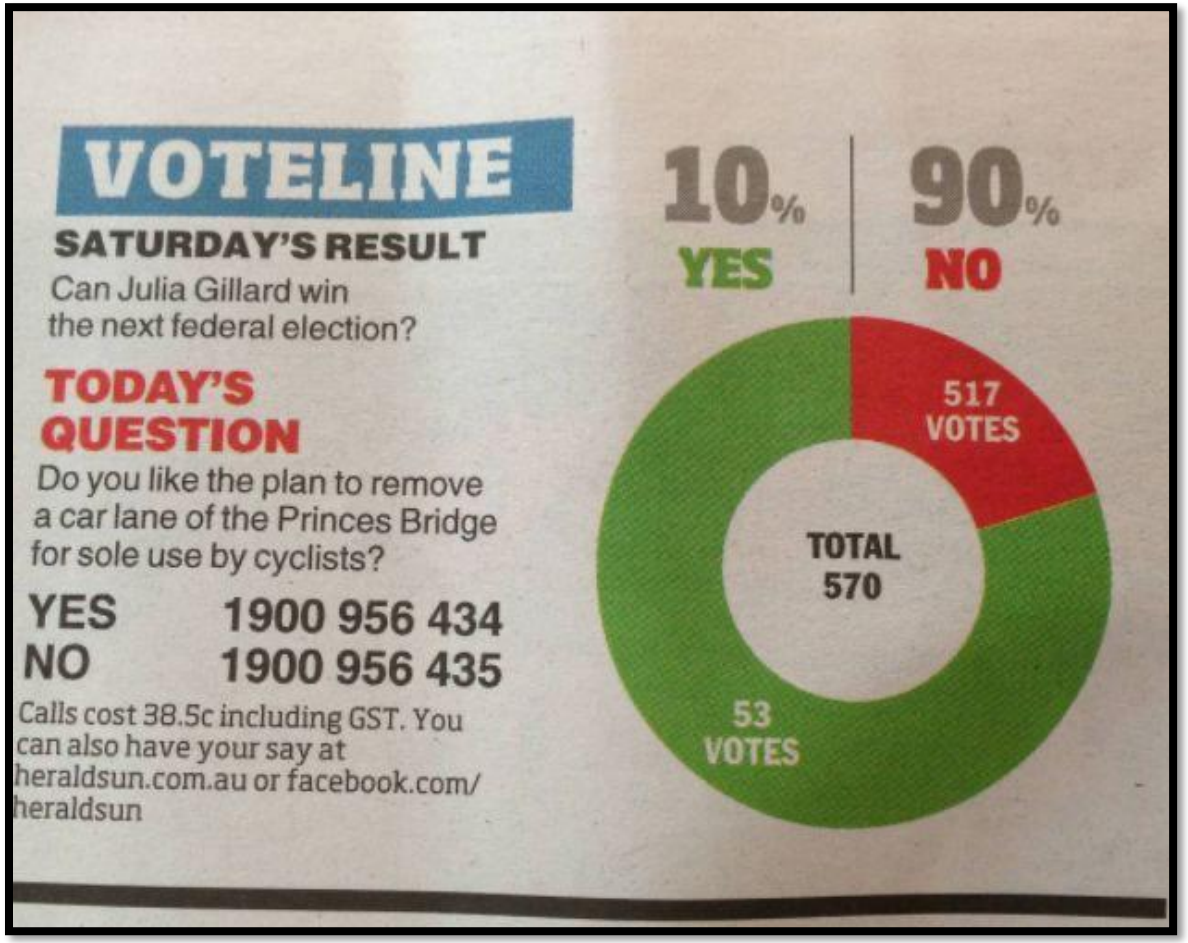
 A. Color

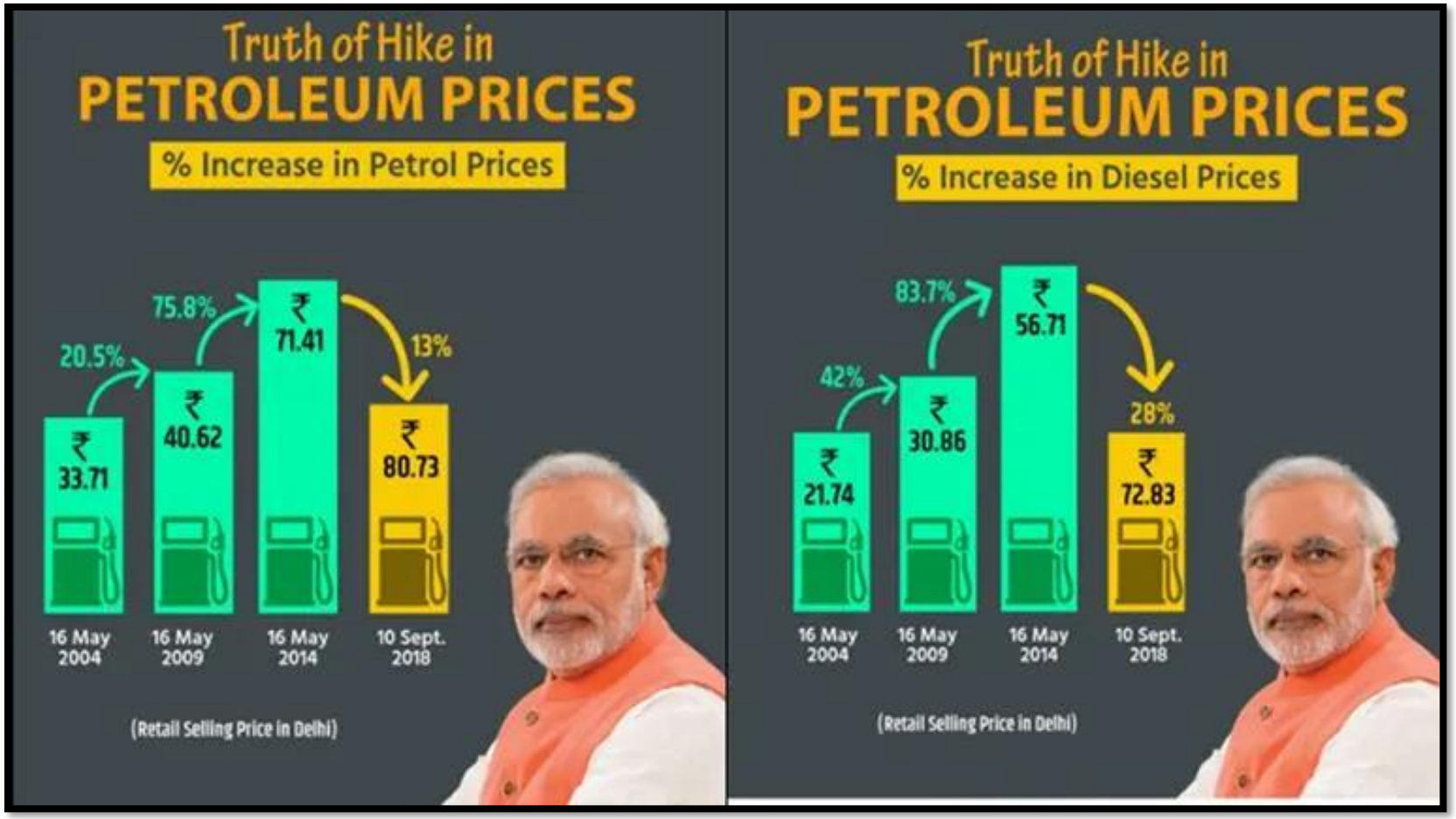
- Colors can evoke **emotions** and **meanings**
  - Red : danger, decline or bad
  - Green : positive, growth
- Poor contrast causes accessibility issues (color blindness, screen quality)
- Use color for **clarity**, not manipulation

Design Choices That Shape Perception

B. Scale and Axes

- Starting Y-axis at non-zero can **exaggerate** trends
- Inconsistent intervals **distort** interpretation
- Logarithmic scales require explanation





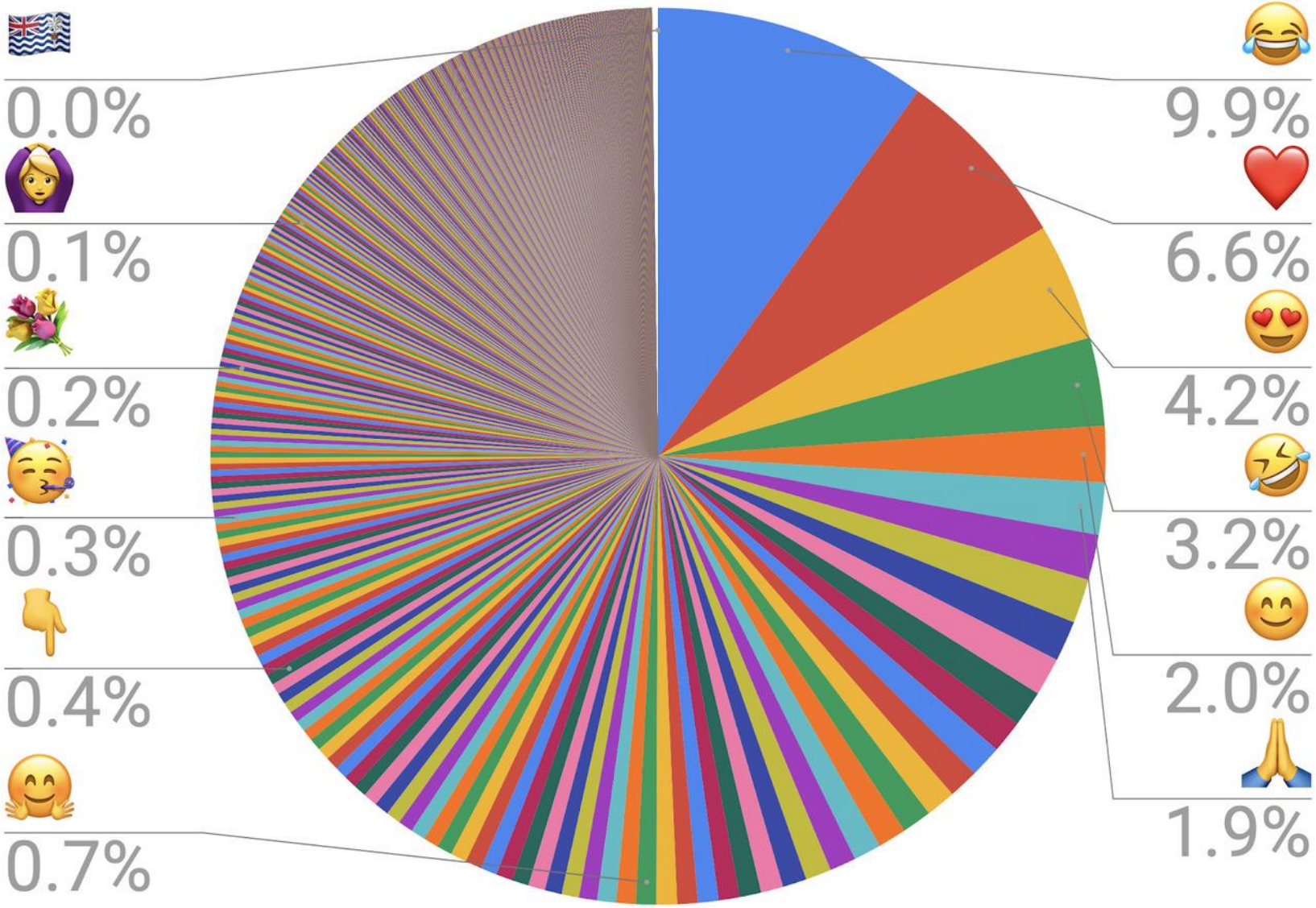
BJP's official twitter handle tweeted out an infographic showing the hike in fuel rates. (Souce: BJP4India/Twitter)

Design Choices That Shape Perception

📊 C. Chart Type

- Using **pie charts** with too many slices causes confusion
- **3D charts** distort angles and bar height
- Use bar/line charts for clarity when appropriate

Good Chart Choice	Misleading Chart Choice
Bar chart for categorical data	3D Pie chart with too many slices
Line chart for time series	Area chart with unclear stacking

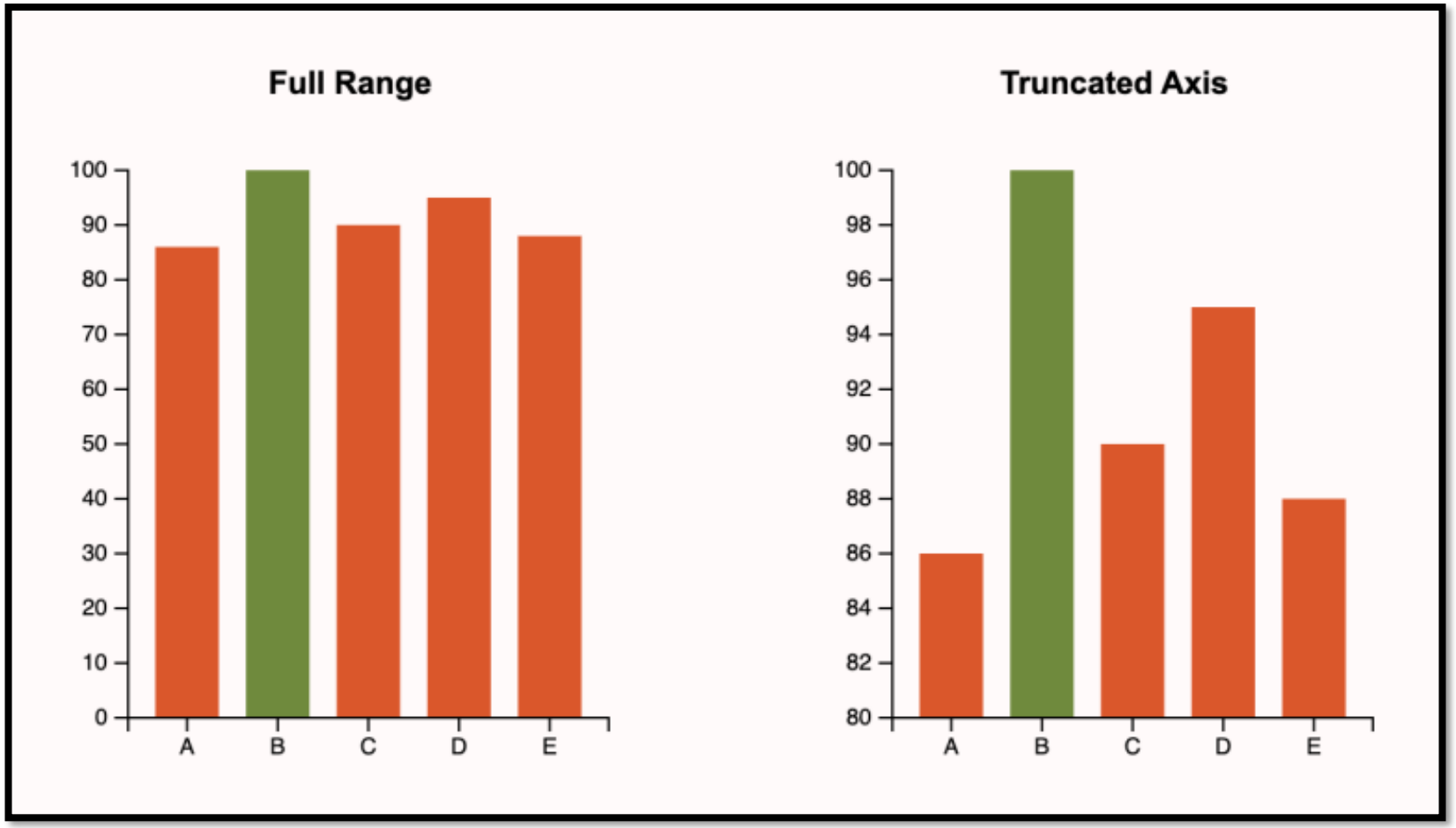

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Cognitive Biases in Data Interpretation

- Even when charts are truthful, viewers interpret them through **mental shortcuts** called cognitive biases.

Bias	Description	Impact
Framing Effect	Perception shifts based on how info is presented	Same data, different emotional response
Anchoring Bias	People rely too heavily on the first piece of info they see	First chart or number skews interpretation
Confirmation Bias	People favor visuals that support their beliefs	Overlook contradictory evidence
Recency Bias	Recent data is perceived as more important	Older patterns ignored

Visualization Comparisons



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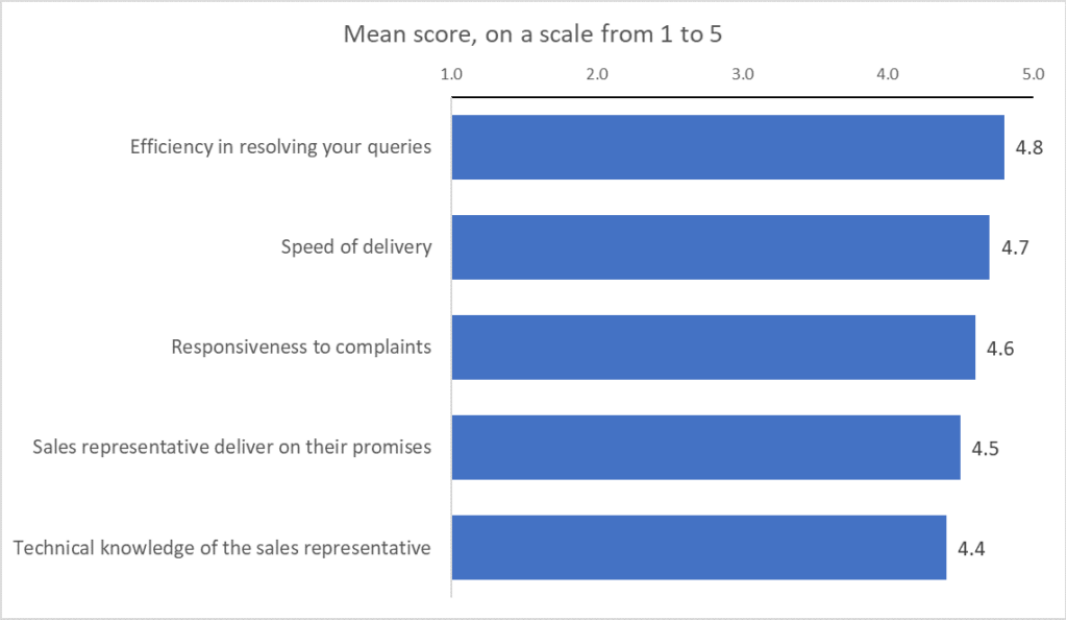
b. Ethics in Practice: Misleading Visual Techniques

c. Understanding Bias in Visualization

A. Common Unethical Practices

a. **Truncated Y-Axes:** Misrepresents data by exaggerating visual difference
b. **Cherry-Picking Data:** Selectively showing only favourable or extreme cases
c. **Ambiguous Legends:** Misleads viewer by hiding or confusing key information
d. **Misused Color Gradients:** Inconsistent color scales distort perceived intensity or value

Cherry-Picking Data



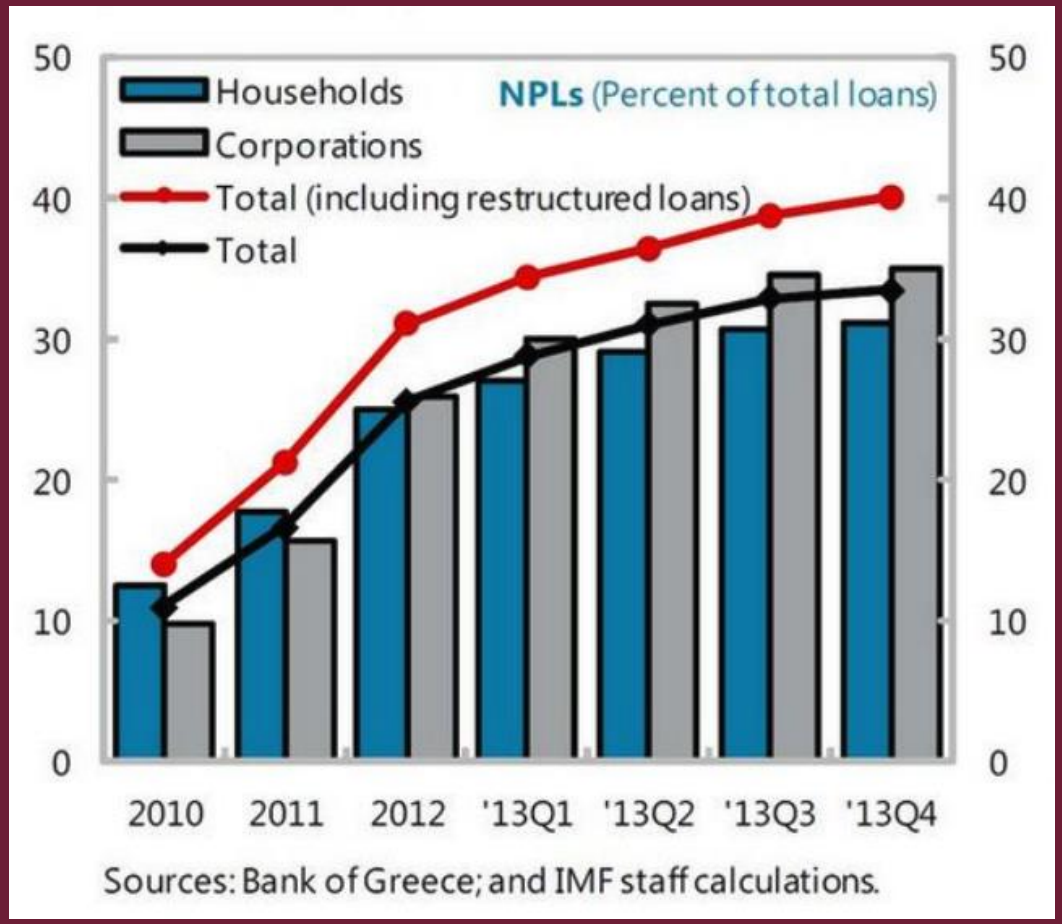


Left: All scores are higher than 4 and the average score would be 4.6, so seems like your customers are happy.

Right: Now your situation doesn't look quite as good, and your average mean score is now 3.9.

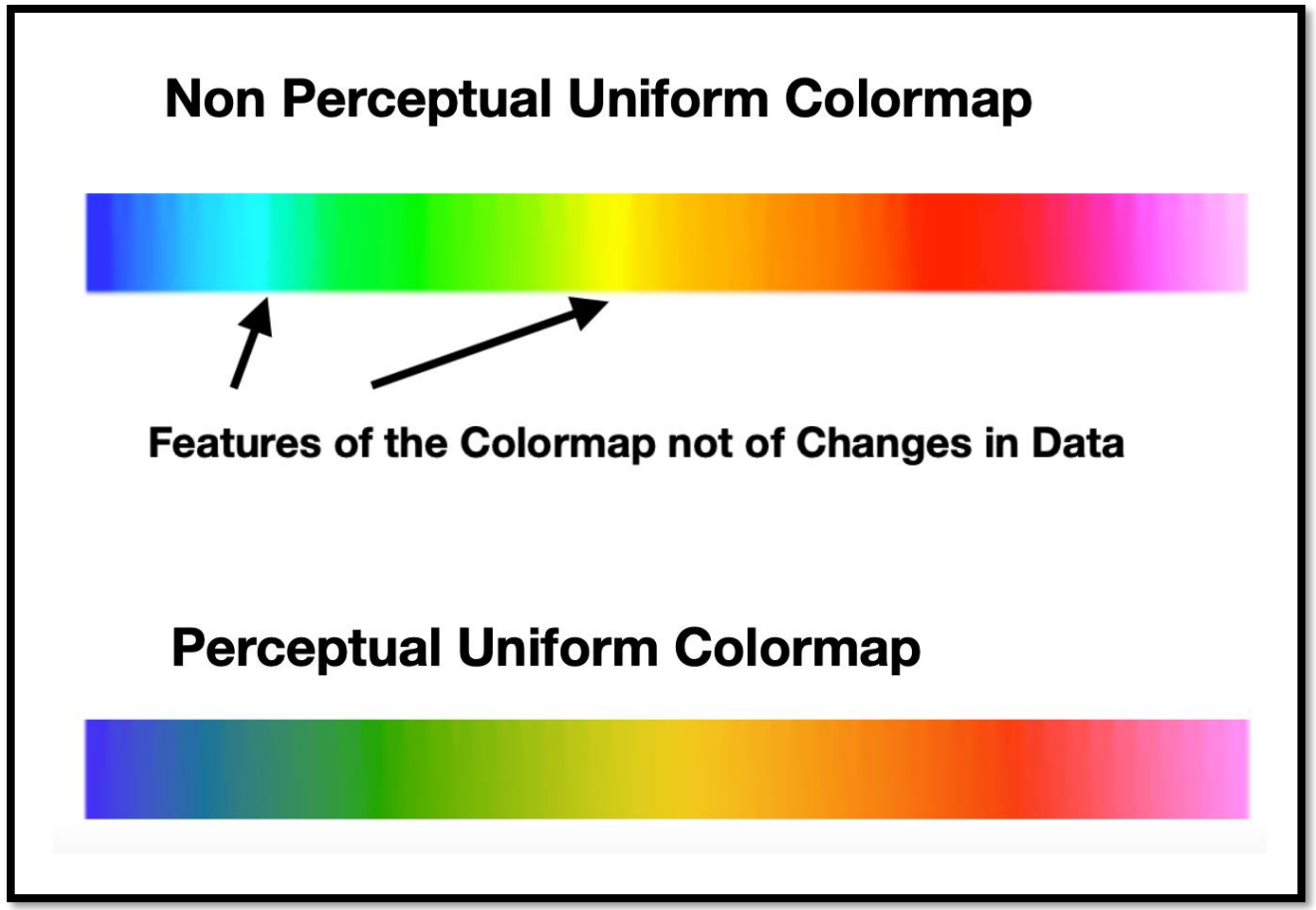
Ambiguous Legends

- The number of non performing loans in Greece is slowing. Look at the first three columns (e.g. 2010, 2011, and 2012) then look at the seventh column.

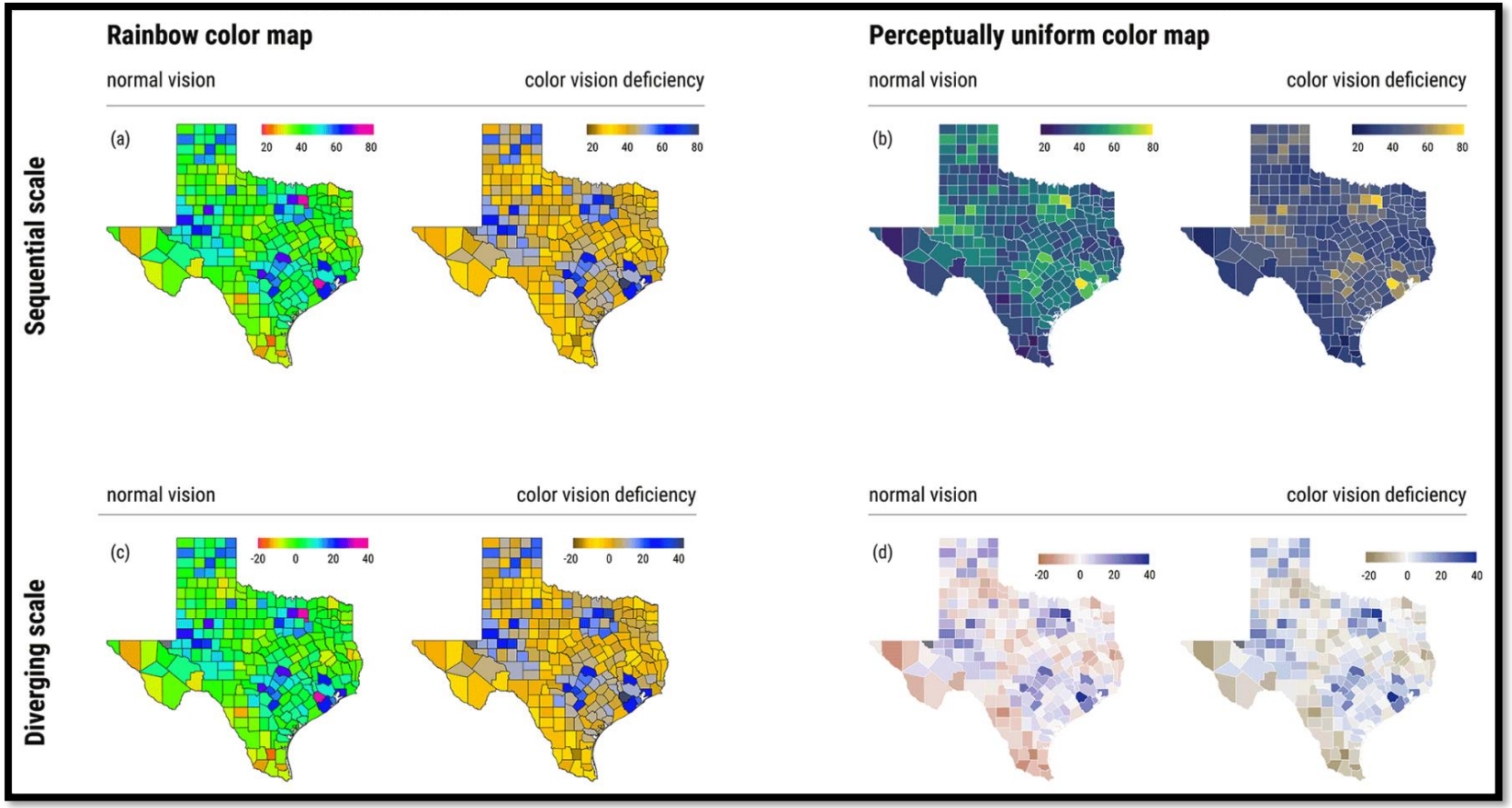


Misused Color Gradients

AVOID this!



Misused Color Gradients



B. How These Techniques Distort Interpretation

- Viewers assume visual accuracy; unethical techniques **manipulate perception**

- Decisions may be made on **false impressions**



C. Visual Integrity

- A concept that emphasizes **honest, accurate, and fair** data presentation

Coined by Edward Tufte:

"Graphical excellence requires telling the truth about the data."

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A. Types of Bias

i. **Explicit Bias**: Intentional manipulation or misrepresentation

ii. **Implicit Bias**: Unconscious choices that influence design and interpretation

B. Sources of Bias

- **Dataset Bias**: Incomplete, outdated, or unrepresentative data
- **Designer Assumptions**: Cultural perspectives, preferred interpretations
- **Audience Expectations**: Preconceived notions and prior knowledge

C. Mitigation Strategies

- **Data Transparency**: Disclose data sources and limitations
- **Inclusive Color Schemes**: Ensure accessibility (e.g., color-blind-friendly palettes)
- **Label Clarity**: Provide clear legends, titles, and annotations
- **Audience-Centered Design**: Consider user context, literacy, and needs

D. Visual Checklist for Ethical Design

- ✅ Does the Y-axis start at zero (when appropriate)?
- ✅ Are labels clear and not misleading?
- ✅ Are colors accessible and consistent?
- ✅ Is all relevant data shown or are exclusions justified?
- ✅ Is the visualization free from exaggeration?
- ✅ Is the audience's context and interpretation considered?



CLASS ACTIVITY



Google Looker Studio Dashboard Design: Avoid These Common Mistakes

https://www.datadashboardhub.com/post/common-mistakes-building-dashboards-in-looker-studio?

KNOWLEDGE CHECK



Achieving excellence in data is commendable, but upholding **integrity** and **honor** in life matters even more.

THANK YOU



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